

| STAT214 Chapter 6 Principal Component Analysis

| 1 Generating Summary Variables: Principal Component Analysis

| 1.1 Dimension Reduction

Dimension reduction is:

- The process of creating a lower-dimensional representation of a dataset
- The goals are
 - to better understand and summarize the strongest patterns and relationships between the variables in the data
 - to make computation on the data easier by reducing its size

| 1.2 Principal Component Analysis

Principal component analysis is a common approach for:

- summarizing low-dimensional linear relationships in high-dimensional datasets
- creating a simplified version of the data that has substantially fewer variables while retaining most of the information of the original dataset
- computing a series of “**orthogonal**” **linear combinations that have the maximum possible variance relative to the origin**

| 1.3 The First Principal Component

The first principal component (PC) is:

- the line along which the projected data points exhibit maximal variation
- the line for which the average perpendicular (projection) distance from each data point to the line is minimized

| 2 Preprocessing: Standardization for Comparability

| 2.1 Standardization

Common preprocessing step for principal component analysis is **Standardization**, which involves

- **mean-centering** (subtracting the mean from each column)
- **SD-scaling** (dividing each column by its standard deviation (SD))

Remark

Neither of these steps are actually required for principal component analysis.

| 2.2 Comparability

There will be some situations in which the relative scale of the variables is meaningful, and scaling them will remove this meaning

Remark

It is more common in practice to apply principal component analysis to mean-centered data.

| 3 Singular Value Decomposition

| 3.1 Singular Value Decomposition

- Principal components are typically computed using **singular value decomposition (SVD)**
- SVD decomposes a data matrix into the product of three matrices: $X = UDV^T$
- The first column of V and the first diagonal entry of D together correspond to the direction and magnitude of the first principal component

| 3.2 The Proportion of Variability Explained

An important result is that **the sum of the variance of the columns in the original dataset equals the sum of the squared singular values**:

$$\sum_j \text{Var}[X_j] = \sum_j d_j^2$$

the proportion of the variability of the original data that is captured or explained by the j th principal component is given by

$$\text{prop of variability explained by PC}_j = \frac{d_j^2}{\sum_i d_i^2}$$

| 3.3 Scree Plot

The **scree plot**:

- is a common method for assessing how many principal components to keep in the reduced dimension dataset
- shows the proportion of variability explained by each principal component either in a bar chart or as a line plot.

| 4 Preprocessing: Gaussianity and Transformations

| 4.1 Gaussianity Assumption

When applying principal component analysis,

- one common assumption is that the variables all follow a theoretical multivariate **Gaussian** (normal) distribution
- one way to check for Gaussianity formally is to use a **Q-Q plot** ("quantile-quantile" plot)

| 4.2 Transformation to Gaussian

- A common transformation for making skewed/asymmetrical data look more Gaussian is the **logarithmic transformation**
- make a judgment call based on whether a log-transformation improves the symmetry of each variable, and later evaluate the impacts of this decision in a PCS stability analysis

Remark

that log-transformations require positive-valued data (or that each variable has a clear lower bound that can be added to each value to artificially make it positive)

| 5 Principal Component Analysis Step-by-Step Summary

1. (**Preprocessing**) Consider whether it makes sense to **mean-center** and/or **SD-scale** the variables
2. (**Preprocessing**) Visualize the distribution of each variable to determine whether a **log-transformation** may increase the symmetry of the distributions.
3. Apply **SVD** to your numeric data matrix to conduct principal component analysis
4. Based on the right-singular vector matrix, V , identify which variables are contributing most to each principal component. Based on the diagonal singular value matrix, D , compute the **proportion of variability explained by each principal component**.
5. Compute the principal component transformation of the original data matrix, X , by multiplying the original data by the right-singular vector matrix, V , as in $X^{PC} = XV$.

| 6 PCS Evaluation of Principal Component Analysis

| 6.1 Predictability

Recompute the summary variables using the **external dataset**, and check that these new principal components are similar to those that were computed.

| 6.2 Stability

| 6.2.1 Stability to Data Perturbations

We typically consider two types of data perturbations:

- sampling uncertainty: using bootstrap data perturbation technique
- measurement uncertainty

| 6.2.2 Stability to Cleaning and Preprocessing Judgment Calls

| 7 Alternatives to Principal Component Analysis

Common alternative approaches include

- nonnegative matrix factorization ([Lee and Seung 1999](#))
- independent component analysis ([Hyvärinen, Karhunen, and Oja 2001](#))
- sparse dictionary learning (sparse coding) ([Olshausen and Field 1997](#))
- kernel principal component analysis ([Schölkopf, Smola, and Müller 1998](#))
- UMAP ([McInnes, Healy, and Melville 2020](#))
- t-SNE ([Van der Maaten and Hinton 2008](#))
- autoencoder ([Goodfellow, Bengio, and Courville 2016](#)).